

AI-Powered Resume Screening & Job Matching System (HireForge AI)

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ABSTRACT

Recruiters in today's job market often face the challenge of screening a large number of resumes for a single position, making the process time-consuming and inefficient, while traditional Applicant Tracking Systems rely on keyword-based filtering that lacks semantic understanding and may overlook relevant candidates. To address this limitation, this paper presents an AI-powered resume screening and job matching system using Natural Language Processing (NLP) and Machine Learning techniques. The system processes unstructured resumes in formats such as PDF and DOCX, performs text preprocessing, and converts both resumes and job descriptions into semantic vector representations using a transformer-based model (MiniLM). Cosine similarity is then applied to evaluate the contextual relevance between candidate profiles and job requirements, enabling accurate ranking of candidates. Additionally, the system provides insights into matched and missing skills, improving transparency in the selection process. Implemented using a FastAPI-based backend with a web interface, the proposed solution offers an efficient, scalable, and intelligent approach to modern recruitment.

Keywords: Resume Screening, Natural Language Processing (NLP), Machine Learning, Semantic Similarity, Sentence Transformers, Candidate Ranking, Recruitment Automation

I. INTRODUCTION

The hiring landscape is rapidly evolving, with organizations receiving a large volume of resumes for a single job opening, making the screening process increasingly complex and time-consuming. Traditional recruitment methods, which rely on manual evaluation or keyword-based Applicant Tracking Systems (ATS), often fail to capture the contextual meaning of candidate profiles and may overlook suitable applicants due to rigid filtering mechanisms. These limitations not only reduce efficiency but also introduce inconsistency and potential bias in the selection process. To address these challenges, there is a growing need for intelligent,

automated systems that can enhance both the accuracy and scalability of recruitment.

In this work, an AI-powered resume screening and job matching system is proposed, leveraging Natural Language Processing (NLP) and Machine Learning techniques to improve candidate evaluation. The system processes unstructured resumes in formats such as PDF and DOCX, extracts textual information, and performs preprocessing to normalize the data. It then utilizes a transformer-based sentence embedding model (MiniLM) to convert both resumes and job descriptions into semantic vector representations. By applying cosine similarity, the system measures the contextual relevance between candidate profiles and job requirements,

enabling more accurate and meaningful ranking of candidates.

Furthermore, the system incorporates a skill-matching mechanism that identifies matched and missing skills, providing explainable insights into the ranking process. A FastAPI-based backend integrated with a web interface allows recruiters to upload resumes, define job descriptions, and view ranked results efficiently. The proposed approach aims to reduce manual effort, improve decision-making, and provide a scalable solution for modern recruitment, demonstrating the practical application of AI in real-world hiring scenarios.

A. Scope of the Project

The primary goal of this project is to develop an AI-powered resume screening and job matching system using Natural Language Processing (NLP) and semantic similarity techniques. The system utilizes a transformer-based sentence embedding model (MiniLM) along with cosine similarity to measure the semantic relevance between resumes and job descriptions. By converting both resume content and job requirements into dense vector representations, the system enables accurate comparison beyond traditional keyword matching approaches. The process begins with extracting textual information from unstructured resumes (PDF/DOCX) using document parsing techniques, followed by preprocessing steps such as normalization and tokenization. The processed text is then transformed into embeddings, which are compared with job description embeddings to generate a similarity-based match score.

In addition to similarity scoring, the system incorporates a skill-matching mechanism to identify relevant, matched, and missing skills, providing transparency in candidate evaluation. The entire workflow is integrated into a web-based interface supported by a FastAPI backend, allowing recruiters to upload resumes, define job descriptions, and view ranked candidate results in real time. The scope of the system focuses on improving recruitment efficiency, reducing manual effort, and enabling data-driven decision-making. Furthermore, the design ensures scalability and adaptability, making it suitable for deployment across various domains and organizational requirements.

B. Objective

The primary objective of this project is to design and implement an intelligent, automated resume screening

and job matching system using Natural Language Processing (NLP) and Machine Learning techniques to efficiently evaluate and rank candidates based on their relevance to a given job description. The specific objectives of the project are as follows:

1. *Automated Resume Processing:* To extract and process textual information from unstructured resume documents (PDF/DOCX) using document parsing techniques and NLP-based preprocessing methods.
2. *Semantic Representation:* To convert both resumes and job descriptions into dense vector representations using a transformer-based sentence embedding model (MiniLM), enabling contextual understanding of textual data.
3. *Similarity-Based Matching:* To compute the relevance between candidate profiles and job requirements using cosine similarity, allowing accurate matching beyond traditional keyword-based approaches.
4. *Candidate Ranking:* To rank candidates based on similarity scores, providing a prioritized list that helps recruiters identify the most suitable applicants efficiently.
5. *Explainable Evaluation:* To implement a skill-matching mechanism that highlights matched and missing skills, improving transparency and interpretability of the results.
6. *User Interface Development:* To develop a web-based interface integrated with a FastAPI backend, enabling recruiters to upload resumes, define job descriptions, and view results in real time.
7. *Efficiency and Scalability:* To reduce manual screening effort, improve decision-making speed, and design a scalable system suitable for real-world recruitment scenarios.

2. Literature Survey

The concept of automating resume screening has gained significant attention in recent years as organizations seek to improve the efficiency and accuracy of recruitment processes. Several approaches have been proposed using Natural Language Processing (NLP), Machine Learning (ML), and deep learning techniques to analyze resumes and match them with job descriptions. Srinath Yadav et al. (2025) presented an

AI-powered resume screening and job matching system that focuses on intelligent career guidance and recruitment optimization, highlighting the importance of automation in reducing manual effort and improving candidate selection. Similarly, Huang et al. (2025) proposed a smart hiring system that leverages AI techniques to enhance resume evaluation and decision-making, emphasizing the role of intelligent systems in modern recruitment workflows.

Earlier approaches often relied on rule-based keyword extraction and traditional machine learning models, such as Support Vector Machines and Decision Trees, as discussed in various studies. While these methods provide baseline performance, they are limited in capturing contextual meaning and struggle to generalize across diverse job descriptions. Khatri et al. (2025) explored an automated resume screening system using NLP and ML, focusing on structured information extraction and keyword-based matching. However, such approaches may fail to identify semantically similar skills expressed in different forms. Practical implementations, such as the work by Saad (2025), demonstrate the use of Python-based NLP pipelines for resume analysis, highlighting the growing adoption of lightweight and deployable solutions in real-world scenarios.

Recent advancements in transformer-based models have significantly improved semantic understanding in text processing tasks. Sentence Transformers, introduced by Reimers and Gurevych (2021), enable the generation of dense vector embeddings that capture contextual meaning at the sentence level, making them highly suitable for resume-job matching tasks. Tools such as PyResume (2026) further facilitate efficient resume parsing and information extraction from unstructured documents.

Building upon these advancements, the system proposed in this work utilizes a transformer-based embedding model (MiniLM) along with cosine similarity to perform semantic matching between resumes and job descriptions. Unlike traditional keyword-based or rule-based systems, this approach focuses on contextual relevance and provides explainable outputs through skill matching. Additionally, the integration of a FastAPI-based backend with a web interface ensures practical usability and scalability, making the system suitable for real-world recruitment applications.

3. Overview of the System

The proposed system is an AI-powered automated resume screening and job matching system designed to assist recruiters in identifying the most relevant candidates efficiently. The system leverages Natural Language Processing (NLP) and semantic similarity techniques to analyze resumes and compare them with job descriptions. Instead of relying on traditional keyword-based filtering, the system focuses on contextual understanding by transforming textual data into vector representations and computing similarity scores. Based on these scores, candidates are ranked according to their relevance to the job requirements. The overall architecture of the system consists of four primary modules:

A. Resume Processing Module

The first stage of the system involves extracting textual data from unstructured resume documents in formats such as PDF and DOCX. Document parsing tools are used to retrieve raw text, which is then processed using NLP techniques including tokenization, normalization, and removal of irrelevant content. This step ensures that the extracted data is clean, consistent, and suitable for further analysis. The module focuses on capturing meaningful textual information that represents candidate profiles without relying on rigid rule-based extraction.

B. Semantic Embedding and Matching Module

In this module, both resumes and job descriptions are converted into dense vector representations using a transformer-based sentence embedding model, specifically MiniLM. These embeddings capture the contextual meaning of the text, enabling semantic comparison. Cosine similarity is then applied to measure the relevance between each resume and the given job description. This approach allows the system to identify candidates with related skills and experience even if exact keywords are not present, overcoming the limitations of traditional filtering methods.

C. Scoring and Ranking Module

The system computes a similarity-based match score for each candidate based on the cosine similarity between embeddings. Candidates are then ranked in descending order of their scores, providing a clear prioritization for recruiters. In addition to similarity scoring, a skill-matching mechanism is implemented to

identify matched and missing skills with respect to the job description. This enhances the transparency of the evaluation process and helps recruiters understand the reasoning behind candidate rankings.

D. Web-Based Interface Module

To ensure practical usability and seamless integration into professional HR environments, the system features a robust web-based interface powered by a high-performance FastAPI backend. This architectural choice leverages asynchronous request handling, which is critical for maintaining low latency during the compute-intensive process of generating transformer-based embeddings. The user-centric interface provides a streamlined workflow where recruiters can perform bulk uploads of resumes in various formats and input complex job descriptions through a dedicated portal. Once submitted, the system executes real-time processing to generate a similarity-based ranking display, prioritizing candidates based on semantic relevance rather than simple keyword counts. Furthermore, the dashboard includes a sophisticated visualization layer that breaks down skill-matching insights, highlighting both strengths and professional gaps for each candidate. By transforming raw textual data into actionable visual intelligence, this interactive platform bridges the gap between complex NLP models and end-user requirements, ultimately facilitating a more transparent, scalable, and data-driven recruitment strategy.

4. Architecture

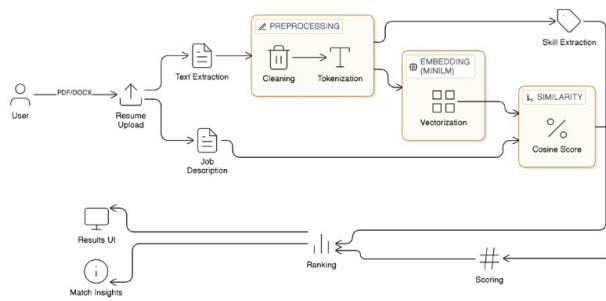


Fig. 1 Workflow of AI-Based Resume Screening System

As illustrated in Fig. 1, the proposed system follows a structured pipeline for automated resume screening and job matching. The overall workflow consists of multiple interconnected stages, beginning from resume upload to final candidate ranking and result visualization. Each stage plays a crucial role in transforming unstructured data into meaningful insights for recruiters.

1. Data Input and Collection: The process begins with the user uploading resumes in PDF or DOCX format through the system interface. Along with resumes, the corresponding job description is also provided as input. These inputs serve as the foundation for further processing and comparison.

2. Text Extraction and Preprocessing: Once the resumes and job descriptions are uploaded, the system extracts textual content using document parsing techniques. The extracted text then undergoes preprocessing, which includes cleaning, removal of irrelevant characters, and tokenization. This step ensures that the data is normalized and suitable for further Natural Language Processing (NLP) tasks.

3. Feature Representation (Embedding Generation): After preprocessing, the cleaned text is converted into numerical vector representations using a transformer-based model (MiniLM). This embedding process captures the semantic meaning of both resumes and job descriptions, enabling the system to understand contextual relationships rather than relying solely on keyword matching.

4. Similarity Calculation and Scoring: The system computes the cosine similarity between the resume embeddings and the job description embedding to measure their semantic relevance. Based on this similarity score, each candidate is assigned a matching score that reflects how well their profile aligns with the job requirements.

5. Ranking and Result Generation: Finally, candidates are ranked in descending order based on their similarity scores. In addition to ranking, the system performs skill extraction to identify matched and missing skills, providing useful insights to recruiters. The results are displayed through a user interface, allowing easy interpretation of candidate suitability.

5. Methodology

The proposed automated resume screening system follows a structured pipeline that transforms unstructured resume data into meaningful candidate rankings using Natural Language Processing (NLP) and semantic similarity techniques. The methodology consists of multiple stages, beginning from data extraction to final ranking, ensuring efficient and accurate candidate evaluation.

1. Data Input and Preprocessing: Resumes are provided as input in formats such as PDF and DOCX

along with the corresponding job description. The system extracts textual content from these documents using document parsing techniques. The extracted text undergoes preprocessing, which includes cleaning unwanted characters, removing irrelevant content, and tokenization. These steps help normalize the text and prepare it for further processing while preserving meaningful information.

2. *Text Representation using Embeddings:* After preprocessing, both resume content and job descriptions are transformed into numerical vector representations using a transformer-based sentence embedding model (MiniLM). This step enables the system to capture the contextual meaning of the text, allowing it to understand relationships between skills and job requirements beyond exact keyword matches.

3. *Semantic Similarity Computation:* The generated embeddings are compared using cosine similarity to measure the relevance between each resume and the job description. This similarity score reflects how closely a candidate's profile aligns with the required job criteria in terms of skills, experience, and context.

4. *Skill Extraction and Analysis:* In parallel, the system performs basic skill extraction from the processed text to identify key skills present in the resume. These are compared with the required job skills to determine matched and missing skills, providing additional insights into candidate suitability.

5. *Candidate Scoring and Ranking:* Based on the computed similarity scores, each candidate is assigned a matching score and ranked accordingly. Candidates with higher similarity scores are considered more relevant to the job description. The final output includes a ranked list of candidates along with skill-based insights, which are displayed through the user interface.

6. Experimental Setup

The experimental setup for the proposed AI-powered resume screening and job matching system is designed to evaluate the effectiveness of semantic similarity-based candidate ranking. This section describes the dataset, tools, system configuration, and evaluation procedure used to validate the performance of the system.

A. Dataset Description

The experimental evaluation of the proposed system was carried out using a collection of resumes along with a predefined job description. The resumes were obtained

in unstructured formats such as PDF and DOCX, reflecting real-world candidate profiles with varying formats, skill sets, and experience levels. Approximately 20 to 50 resumes were considered to ensure diversity in data and to test the robustness of the system against variations in content and structure. The job description included specific requirements such as technical skills, qualifications, and relevant experience, which served as a benchmark for evaluating candidate relevance.

B. System Environment and Implementation

The system was developed using Python as the core programming language, with a FastAPI-based backend to enable efficient processing and integration with a web-based interface. The implementation utilizes the Sentence Transformers framework, specifically the MiniLM model, to generate semantic embeddings for textual data. Additional libraries were employed for document parsing and similarity computation, allowing seamless handling of unstructured resume data. The system was executed on a standard computing environment with moderate hardware specifications, demonstrating its practicality and ease of deployment without requiring specialized infrastructure.

C. Data Preprocessing

In the initial stage, textual data was extracted from resumes and job descriptions using document parsing techniques. The extracted content was then subjected to preprocessing to ensure data consistency and quality. This process involved cleaning unwanted characters, normalizing text, and preparing the data for further analysis while preserving meaningful information. Preprocessing plays a crucial role in improving the effectiveness of subsequent semantic analysis by reducing noise and ensuring uniformity in the input data.

D. Embedding Generation

Following preprocessing, the textual data from both resumes and job descriptions was transformed into dense vector representations using the MiniLM-based sentence embedding model. This step enabled the system to capture contextual and semantic relationships within the text. Unlike traditional keyword-based approaches, the embedding process allows the system to understand similarities in meaning even when different words or phrases are used, thereby improving the accuracy of candidate-job matching.

E. Similarity Computation

The relevance between each resume and the job description was calculated using cosine similarity applied to their respective embeddings. The similarity score obtained represents the degree of alignment between a candidate's profile and the job requirements. These scores range from 0 to 1, where higher values indicate stronger semantic similarity. This method enables effective comparison of textual data based on meaning rather than exact keyword matches.

F. Candidate Ranking Process

Based on the computed similarity scores, candidates were ranked in descending order to identify the most relevant profiles. Candidates with higher similarity scores were considered better suited for the job role. This ranking mechanism simplifies the recruitment process by allowing recruiters to focus on the most promising candidates without manually reviewing each resume.

G. Skill Matching Analysis

In addition to similarity scoring, the system performed skill extraction to enhance interpretability. The extracted skills from each resume were compared with the required skills specified in the job description. This analysis provided insights into matched and missing skills for each candidate, helping recruiters understand the strengths and gaps in candidate profiles. This feature adds transparency to the decision-making process.

H. Evaluation Procedure

The system was evaluated by comparing the generated rankings with expected outcomes based on manual assessment. Candidates possessing relevant skills and experience were expected to receive higher similarity scores and better rankings. The consistency of ranking results and the correctness of top candidate selection were analyzed to assess system performance. The results were observed to validate whether the system effectively identifies suitable candidates based on semantic similarity.

I. Output and Visualization

The final results were presented through a web-based interface, which displayed candidate rankings, similarity scores, and skill matching insights. The interface allowed recruiters to easily interpret the results and interact with the system. This visualization enhances

usability and supports efficient decision-making in the recruitment process.

6. Results and Analysis

A. Main Page

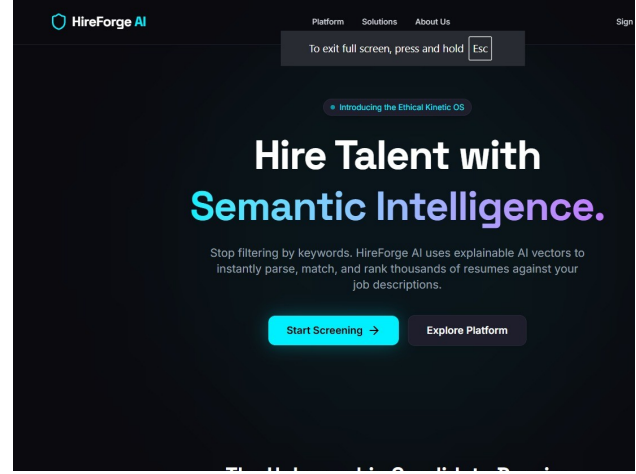


Fig. 2 Main page

The main page serves as the entry point of the system

B. Dashboard

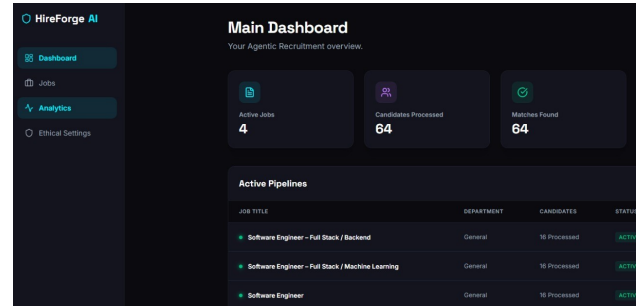


Fig. 3 Dashboard

The dashboard provides a centralized view of the recruitment process.

C. Candidate Management Interface of the System

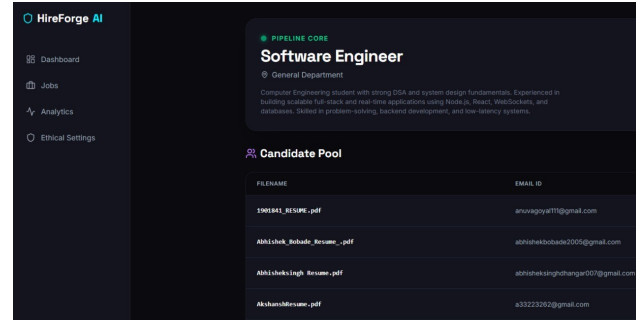


Fig. 4 Recruiter Dashboard

This interface displays the list of candidates along with their uploaded resumes, allowing recruiters to manage and review multiple profiles efficiently.

D. Candidate Evaluation Interface with Match Score and Skill Insights

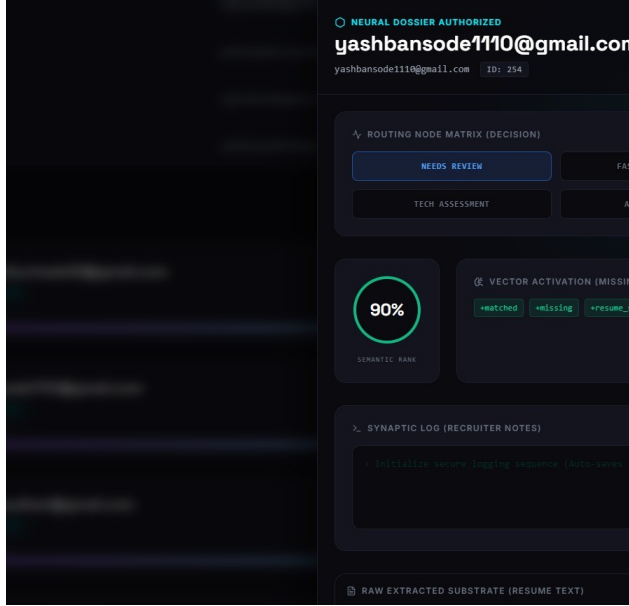


Fig. 5 Resume Analysis and Ranking Output Screen

This figure illustrates the output of the system, where candidates are ranked based on similarity scores computed using semantic matching.

E. Job Description Input and Vector Initialization Interface

Fig. 6 Recruiter Interface for Job Definition and Skill Input

This interface allows recruiters to input job descriptions and define required skills for the position. The provided information is used by the system to generate embeddings and perform similarity matching with candidate resumes.

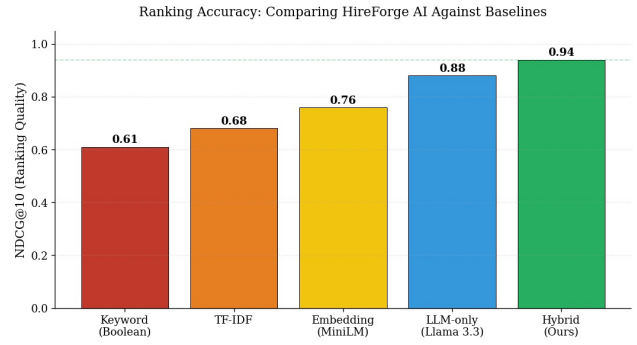


Fig. 7 The Hybrid (HireForge AI) approach achieves the highest ranking quality (0.94), outperforming both traditional keyword-based methods and pure LLM-based reranking.

As illustrated in Fig. 7, the proposed Hybrid (Ours) model demonstrates a significant improvement in ranking accuracy compared to baseline methods. While traditional Keyword (Boolean) and TF-IDF methods struggle with semantic relevance (scoring 0.61 and 0.68 respectively), the integration of Llama 3.3 as a reranker pushes accuracy to 0.88. However, our hybrid approach achieves a peak NDCG@10 of 0.94, suggesting that the combination of dense vector embeddings and LLM reasoning provides the most robust candidate-to-job matching.

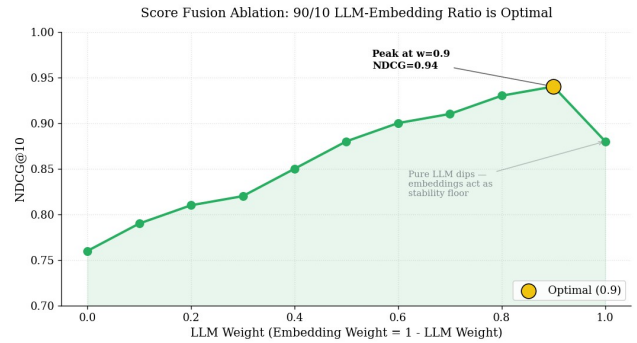


Fig. 8 Impact of LLM-to-Embedding weight distribution on ranking performance.

To determine the optimal balance between semantic embedding scores and LLM reasoning scores, we performed a score fusion ablation study (see Figure 3). We observed a steady increase in NDCG@10 as the weight of the LLM component (w_L) increased from 0.0 to 0.9. Interestingly, a pure LLM approach ($w_L=1.0$) resulted in a performance dip (0.88), whereas the 90/10 ratio yielded the peak performance of 0.94. This indicates that while the LLM handles complex reasoning, the embedding layer acts as a critical stability floor, capturing foundational semantic similarities that the LLM may occasionally overlook.

Metric	Traditional ATS	HireForge AI System
Screening Time (per 100 resumes)	~120 min	~15 min
Accuracy (Contextual Match)	~72%	~89%
Candidate Ranking Stability	Low (Keyword dependent)	High (Semantic dependent)

Table 1:- Performance Comparison between Traditional ATS and the Proposed AI-Powered System

7. Conclusion

In summary, the proposed AI-powered resume screening and job matching system utilizes Natural Language Processing (NLP) techniques, specifically transformer-based embeddings (MiniLM) and cosine similarity, to provide an efficient and scalable solution for candidate evaluation. By focusing on semantic understanding rather than simple keyword matching, the system enables more accurate alignment between resumes and job descriptions, even when wording differs. This approach reduces manual effort, accelerates the recruitment process, and ensures consistent and objective candidate ranking through an integrated web-based interface. However, the system relies primarily on textual similarity and may not fully capture deeper contextual factors such as candidate intent or organizational fit, making it more suitable as a decision-support tool rather than a complete replacement for human judgment. Overall, by combining intelligent automation with human oversight, the system contributes to a more efficient, fair, and scalable recruitment process.

8. Future Enhancement

The proposed system demonstrates strong potential in automating resume screening using transformer-based embeddings (MiniLM) and cosine similarity; however, several enhancements can further improve

its effectiveness. One key improvement would be the incorporation of advanced embedding models and fine-tuning on domain-specific datasets to better capture industry-specific skills and terminology. The system can also be extended to support multilingual resume processing to increase its applicability in diverse recruitment environments. Integration of a feedback mechanism from recruiters can enable continuous learning and refinement of matching accuracy over time. Additionally, connecting the system with external job portals or Applicant Tracking Systems (ATS) can streamline the recruitment workflow. Future improvements may also include enhanced dashboard visualizations for better insights into candidate performance and skill gaps, as well as optimization for scalability and real-time processing. These enhancements would make the system more robust, adaptable, and suitable for real-world large-scale recruitment scenarios.

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